Shear Wave Propagation in the Brain

A Finite Element Analysis in COMSOL Multiphysics

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Student Contributions

This project was a collaborative effort, with each team member developing and analysing one finite-element (FE) model independently while maintaining a shared workflow. Valeria was responsible for the three-dimensional cube model, Giovanni for the two-dimensional square model, and Lisa for the two-dimensional circle model. Each model was implemented, simulated and iteratively refined by its respective developer, enabling direct comparison of geometric effects within a consistent modelling framework.

Project coordination and technical alignment were ensured through weekly meetings with the supervisor. In these meetings, the results obtained during the preceding week were presented and critically discussed, and unresolved methodological or numerical issues were addressed collectively. Based on these discussions, new tasks and priorities for the subsequent week were defined, allowing the group to maintain a coherent direction while preserving the independence of the modelling work.

In the later stages of the project, workload was distributed according to the demands of the remaining tasks. Giovanni took primary responsibility for designing and executing the parametric sensitivity study, working in close collaboration with Valeria, who contributed to the parameter sweep analysis alongside the continued refinement of her three-dimensional cube model. Lisa focused on theoretical research, consolidating and extending the physical and mathematical background underlying the simulations, while simultaneously continuing the finalisation of her two-dimensional circle model.

The handling of model-specific challenges was primarily undertaken by the respective contributor. Valeria's cube simulations were computationally demanding and therefore required substantial time for parameter tuning and repeated runs. Conversely, Lisa encountered geometry-driven numerical artefacts in the circle model, including increased boundary effects, which required independent investigation.

The project combined individual responsibility for modelling tasks with weekly integration to ensure the comparability of results and consistent methodological documentation across the three geometries.

Abstract

Magnetic Resonance Elastography (MRE) relies on shear-wave motion to infer tissue stiffness, but in brain-like, fluid-saturated media the wave field can be strongly affected by geometry, fluid–solid coupling, and boundary conditions, making standard plane-wave inversion assumptions unreliable. In this project, controlled finite element (FE) simulations are developed in COMSOL Multiphysics to study steady-state shear wave propagation in a brain-like poroelastic material and to quantify how domain geometry, boundary treatment, and material parameters affect wave-field quality and spatial attenuation characteristics. Brain tissue is modelled using Biot poroelasticity, in which solid displacement and pore pressure are fully coupled and multiple wave modes can arise; the shear-dominant motion of the solid frame constitutes the primary signal of interest.

Three benchmark geometries are implemented to isolate dimensional and boundary effects: a 3D cube, a 2D square, and a 2D circle. Each model is excited by a prescribed harmonic displacement, and frequency-domain sweeps from 10–100 Hz, covering the clinically relevant brain MRE excitation range. Reflective or free boundaries introduce strong backscattering and standing-wave interference in both displacement and pore-pressure fields, producing spatial artefacts that obscure the propagating wave structure. In contrast, periodic lateral boundary conditions and Perfectly Matched Layers (PMLs) substantially reduce reflected energy, yielding clearer wave fronts and spatially coherent 1D displacement profiles. For the circular geometry, an annular PML paired with a sufficiently refined interior mesh is required to suppress numerical artefacts at the curved domain–PML interface.

Quantitative comparison across the final model configurations reveals close agreement between the rectilinear geometries: the 2D square and 3D cube exhibit attenuation coefficients differing by only 5.2% and phase velocities of approximately 1.9–2.0 $\mathrm{m\,s^{-1}}$ at 50 Hz, consistent with experimentally reported brain tissue stiffness values. The circular geometry exhibits nearly double the attenuation coefficient and a penetration depth approximately 50% shorter than the rectilinear models, attributable to geometric scattering rather than differences in material dissipation. A parametric sensitivity analysis identifies hydraulic permeability as the dominant material parameter governing displacement magnitude and energy dissipation across all geometries, with its influence increasing at higher excitation frequencies.

These results establish a reproducible FE framework for systematically evaluating when simplified MRE wave assumptions are credible in poroelastic brain-like media and for identifying boundary treatments that yield interpretable wave fields suitable for subsequent stiffness inference.

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1 Introduction

The human brain is among the most mechanically complex organs in the body, and its structural integrity is increasingly recognised as a sensitive indicator of neurological health. Conditions such as Alzheimer's disease, multiple sclerosis, hydrocephalus, and intracranial tumours are all associated with measurable changes in brain tissue stiffness, yet conventional anatomical MRI cannot detect the mechanical alterations that often precede visible morphological changes [1, 2, 3]. Magnetic Resonance Elastography (MRE) is a non-invasive imaging technique that estimates brain stiffness in vivo by introducing shear waves into the tissue through skull vibration, measuring the resulting micron-scale displacements in MRI phase images, and reconstructing spatially resolved stiffness maps by working backwards from the measured wave field to the underlying tissue properties [4, 5]. Because the procedure is entirely non-invasive and repeatable, it holds considerable promise for disease diagnosis, staging, and monitoring of treatment response.

The accuracy of these stiffness maps depends critically on the wave model assumed during reconstruction. Most approaches treat shear waves as plane waves travelling undisturbed through a homogeneous, unbounded medium, so that tissue stiffness can be estimated directly from the local wavelength [5]. The physiological reality of the brain, however, departs substantially from these assumptions. Brain tissue is geometrically finite, internally heterogeneous, and characterised by a fluid-saturated porous microstructure rather than a single-phase elastic solid. The mechanical response is therefore governed by fluid–solid coupling: as the solid matrix deforms, the pressurised pore fluid resists and redistributes, altering both the phase velocity and the attenuation of propagating waves in ways that single-phase continuum models cannot capture [6]. Furthermore, at the excitation frequencies typical of clinical brain MRE protocols (25–80 Hz), the shear wavelength becomes comparable to the dimensions of many brain structures, so that waves reflected from tissue boundaries and internal interfaces actively interfere with the propagating field and introduce standing-wave patterns that cannot be treated as negligible artefacts. Such boundary effects (specular reflections, standing-wave interference, and phase distortion) can bias stiffness estimates directly, since the apparent local wavelength extracted from the distorted field no longer corresponds to the true tissue shear modulus.

A key difficulty is that in a physical MRE experiment, geometry, material properties, and boundary effects all act simultaneously and cannot be disentangled from one another. Identifying which factors drive errors in the predicted stiffness maps, and under what conditions simplified assumptions remain credible, requires a setting where each variable can be independently controlled. Finite element (FE) simulation provides exactly this, offering a reproducible computational environment where geometry, excitation, material parameters, and boundary conditions can all be specified precisely and varied systematically.

1.1 Objectives

The goal of this project is to obtain a better understanding of shear wave propagation through brain tissue using finite element simulations implemented in COMSOL Multiphysics [7]. Three benchmark geometries are studied, a three-dimensional cube, a two-dimensional square, and a two-dimensional circle, with the general aim of identifying under what conditions the simplified assumptions underlying standard MRE wave models remain physically credible. The specific research questions are:

1. How does the choice of boundary condition affect the wave field within a given domain geometry, in terms of reflected-energy suppression and the spatial regularity of displacement profiles?
2. How do wave propagation characteristics (spatial wavelength, attenuation rate, and frequency response) differ between the three domain geometries under otherwise identical boundary treatments, and to what extent can these differences be attributed to geometric shape rather than material behaviour?
3. How do selected material parameters alter shear wave propagation characteristics across the simulated frequency range?

The primary outcome is a comparative analysis of simulated displacement across the three domain geometries (square, circle, cube) and boundary treatments, alongside an assessment of how material parameters influence the simulated wave fields in each domain geometry.

2 Literature Review

2.1 Principles of Magnetic Resonance Elastography

Magnetic resonance elastography (MRE) is a non-invasive imaging technique that measures tissue mechanical properties by visualising the propagation of

externally induced shear waves with MRI. In a typical MRE experiment, harmonic vibrations generated by an external actuator produce micron-scale displacements within the tissue, which are encoded in the phase of the MR signal using motion-sensitising gradients. These displacement fields are subsequently processed with inversion algorithms to reconstruct spatial maps of tissue stiffness or shear modulus. A widely used early inversion approach is local frequency estimation (LFE), which estimates local stiffness from the measured spatial wavelength under the assumptions of locally homogeneous material behaviour and approximately plane-wave propagation in an unbounded medium. This methodological framework established the basis of quantitative MRE and enabled its application to the brain, while later studies also emphasised that the accuracy and physical interpretability of reconstructed parameters depend strongly on whether these simplifying assumptions are appropriate for the tissue and wave-propagation conditions under investigation [4, 5, 8, 6].

2.2 From Clinical Observation to Computational Modelling

The clinical motivation for brain MRE rests on a consistent body of evidence linking tissue stiffness to neurological pathology. Wuerfel et al. [1] demonstrated that MRE could detect tissue softening in multiple sclerosis patients consistent with inflammatory demyelination, a change invisible to conventional structural MRI. Murphy et al. [2] showed that preoperative MRE-derived stiffness of meningiomas correlated with surgical difficulty, offering a practical planning tool. Hiscox et al. [3] identified viscoelastic differences in subcortical grey matter between young and older healthy adults, establishing stiffness as a marker of normal ageing as well as pathology. Across these studies, a recurring finding is that stiffness changes are spatially heterogeneous and anatomically confined, placing demands on inversion accuracy that become increasingly hard to satisfy as the assumptions of the wave model depart from the physical reality of the brain.

The inversion pipeline underpinning these clinical studies was formalised by Manduca et al. [5], whose local frequency estimation approach extracts wavelength from the measured displacement field and converts it to stiffness under the assumption of plane wave propagation in a locally homogeneous, isotropic medium. Johnson et al. [8] and Weaver et al. [9] extended this to whole-brain acquisition protocols operating in the 25–80 Hz range, a frequency window chosen to balance skull transmission, tissue penetration, and spatial resolution of the stiffness map. In the present work, the simulation sweep is extended to 90 Hz to include the common clinical band while probing higher frequencies where wavelengths shorten and sensitivity to boundary treatment and geometry increases.

2.3 Material Modelling of Brain Tissue

The appropriateness of the wave model depends on how accurately the underlying material description captures brain tissue behaviour. Budday et al. [10] performed the most comprehensive mechanical characterisation of human brain tissue to date, testing multiple deformation modes and demonstrating strong nonlinearity, pronounced conditioning effects, and significant regional stiffness variation. Their results showed that no single linear elastic modulus adequately describes the full mechanical response, and that the high water content and biphasic microstructure of the tissue make a poroelastic treatment physically well motivated.

McGarry et al. [6] provided the theoretical argument for adopting poroelasticity specifically in the MRE inversion context. They showed that while both poroelastic and viscoelastic models are uniquely invertible at low MRE frequencies, the poroelastic formulation retains this property in regimes where viscoelastic inversion becomes degenerate. In such regimes, the inverse problem ceases to admit a unique solution, meaning multiple stiffness distributions may produce an identical measured wave field. Importantly for this project, McGarry et al. [6] demonstrate that poroelastic forward simulations require both solid displacement and fluid pressure or flux to be specified at every boundary simultaneously. Omitting either field renders the system mathematically underdetermined. This produces spurious wave-field artefacts that are indistinguishable from genuine material effects. This finding directly motivated the iterative boundary condition strategy developed in this work.

2.4 Numerical Methods and Boundary Treatment

The finite element implementation of poroelastic wave equations was placed on a rigorous footing by Atalla et al. [11] and Debergue et al. [12], who developed the mixed $(\mathbf{u}, p_f)$ formulation that reduces the coupled solid–fluid problem to a displacement–pressure system suitable for standard FE discretisation. This formulation is the basis of the COMSOL Poroelastic Waves interface [13] used in this project.

The treatment of absorbing boundaries in wave simulation was fundamentally advanced by Berenger [14], who introduced the Perfectly Matched Layer as a theoretically reflectionless absorbing region based on complex coordinate stretching. The PML was originally introduced for electromagnetic wave problems [14] and has since been extended to elastic and poroelastic wave equations [15], becoming a widely adopted approach for suppressing boundary reflections in computational wave domains. Its performance in practice depends on parameterisation, and in soft, highly attenuating media such as brain tissue, the curvature of the stretching function requires careful tuning to avoid residual interface reflections, as demonstrated in the boundary condition development of this project.

2.5 Research Gap and Motivation

To the best of our knowledge, no controlled computational study has systematically isolated the contributions of geometry, boundary conditions, and material parameters to wave-field quality in brain-like poroelastic media. Existing FE studies have focused on inversion validation in realistic brain geometries [6], where individual contributions cannot be separated. The validity of the plane-wave and the assumption of a homogeneous, reflection-free unbounded medium that underpin standard MRE inversion has not been directly tested in a poroelastic framework across the clinical frequency range, and no benchmark dataset of controlled simulations exists to support such an evaluation.

3 Theoretical Background

3.1 Biot's Theory of Poroelastic Wave Propagation

Brain tissue can be modelled as a fluid-saturated porous solid, where the mechanical response depends on both the deformation of the solid matrix and the motion of the fluid within the pores. Biot's theory [16, 17, 18] provides the governing equations for the coupled solid–fluid system. Accordingly, the material response is modelled as linear, isotropic, and homogeneous, and the small-strain assumption is adopted throughout.

The theory rests on two constitutive relations. The first states that the total stress in the material is the sum of the stress carried by the drained solid skeleton and the contribution from the fluid pore pressure p_f,

$$\boldsymbol{\sigma} = \mathbf{C}\,\boldsymbol{\varepsilon} - \alpha_B\, p_f\, \mathbf{I}, \tag{1}$$

where $\mathbf{C}$ is the drained elasticity tensor, $\boldsymbol{\varepsilon}$ is the solid strain tensor, $\mathbf{I}$ is the identity tensor, and α_B is the Biot–Willis coefficient [16]. This coefficient quantifies how much of any applied volumetric load is carried by the pore fluid. For a very soft matrix such as brain tissue, $\alpha_B \approx 1$, meaning the fluid carries essentially all of the volumetric stress.

The second relation links changes in pore pressure to changes in solid volume through the Biot modulus M,

$$\zeta = \alpha_B\, \nabla \cdot \mathbf{u} + \frac{p_f}{M}, \tag{2}$$

where ζ is the increment in fluid content and $\mathbf{u}$ is the solid displacement vector. The Biot modulus M controls the storage capacity of the medium, that is, how much fluid can enter or leave for a given pressure change, and is determined by the porosity ε_p, the solid grain bulk modulus K_s, and the fluid bulk modulus K_f:

$$\frac{1}{M} = \frac{\alpha_B - \varepsilon_p}{K_s} + \frac{\varepsilon_p}{K_f} \tag{3}$$

where K_s is the intrinsic bulk modulus of the solid grains, independent of the pore structure, and K_f is the bulk modulus of the pore fluid. For brain tissue, K_f is approximated from the fluid compressibility χ_f as $K_f = 1/\chi_f$, with the value given in Table 2.

Fluid transport within the pore space follows Darcy's law, in which fluid flux is proportional to the pore pressure gradient and inversely proportional to the ratio of fluid viscosity η_f to permeability κ. Low permeability or high viscosity slows fluid redistribution, increasing coupling between solid and fluid motion and enhancing wave attenuation.

Combining these relations with Newton's second law for the solid and the fluid continuity equation gives Biot's wave equations. Specifically, Newton's second law for the solid skeleton and the fluid continuity equation give, in the frequency domain:

$$\nabla \cdot \sigma + \omega^2 \rho_{av}\, \mathbf{u} = 0 \tag{4}$$

$$-i\omega\zeta + \nabla \cdot \left(\frac{\kappa}{\eta_f} \nabla p_f \right) + i\omega\alpha_B\, \nabla \cdot \mathbf{u} = 0 \tag{5}$$

where ω is the angular frequency [18, 11]. Together with the constitutive relations 1 and 2, these form the dynamic Biot wave equations [18] solved by COMSOL's Poroelastic Waves interface [19].

The material is modelled as a single homogeneous poroelastic medium, all constitutive and fluid parameters used in the simulations are defined and summarised in Table 2. This mixed $(\mathbf{u}, p_f)$ formulation

requires only one additional scalar variable per node rather than a full vector fluid displacement field, making it numerically efficient for FE implementation.

3.2 Wave Modes

Biot's theory predicts three wave modes [20]: two compressional and one shear. The fast compressional wave involves solid and fluid moving nearly in phase and propagates at speeds close to those of the drained solid. The slow compressional wave, which involves out-of-phase motion between the solid and fluid phases, is strongly attenuated by viscous dissipation and is therefore not expected to propagate appreciably at the frequencies and material properties considered here. The shear wave is the mode of interest for MRE. Because shear deformation produces no volume change in an isotropic medium, it does not drive bulk fluid flow, and the shear wave therefore propagates largely independently of permeability and fluid compressibility. However, fluid-solid coupling does introduce a degree of viscous attenuation and a minor modification of the shear wave speed [17]. Its speed is governed by the drained shear modulus G_d and the average saturated density,

$$c_s = \sqrt{\frac{G_d}{\rho_{\mathrm{av}}}}, \qquad \rho_{\mathrm{av}} = (1-\varepsilon_p)\,\rho_s + \varepsilon_p\,\rho_f, \quad (6)$$

providing the direct link between the simulated wave field and the tissue stiffness that MRE aims to recover.

3.3 Boundary Conditions for Poroelastic Waves

In a poroelastic model, boundary conditions must be specified simultaneously for both the solid displacement and the fluid field. A prescribed displacement condition enforces a specific solid motion on the boundary. A zero fluid flux condition $Q = 0$ enforces an impermeable wall. In COMSOL notation, Q denotes the normal filtration flux, so $Q = 0$ implies that no pore fluid crosses the boundary [19]. A drained boundary represented by $p_f = 0$ allows pore pressure to equilibrate with the surroundings. As shown by McGarry et al. [6], consistent specification of both fields at every boundary is essential for a well-posed problem.

Periodic boundary conditions enforce identical displacement and pressure fields on opposing face pairs, replicating the domain infinitely in those directions and eliminating reflections from lateral boundaries. This approximates propagation through a laterally unbounded medium, which is a common assumption in MRE forward modelling [8].

A Perfectly Matched Layer (PML) [14] is an absorbing region in which complex-valued coordinate stretching causes outgoing waves to decay exponentially without generating interface reflections, regardless of frequency or angle of incidence. In COMSOL's Poroelastic Waves module, the PML absorbs both solid and fluid wave components simultaneously [13]. An important practical constraint is that PML and periodic conditions cannot be applied to the same boundary simultaneously, as the PML requires outgoing waves to be absorbed while the periodic condition imposes a relationship between two opposing faces.

3.4 Finite Element Method and Spatial Discretisation

A frequency-domain formulation directly yields the steady-state complex amplitude of $(\mathbf{u}, p_f)$ at each excitation frequency, matching the harmonic nature of MRE actuation. The governing equations are cast in weak integral form and solved as a global algebraic system at each frequency step [11, 12]. The two-dimensional geometries are treated as plane-strain representations of the three-dimensional problem, enabling isolation of geometric effects at a fixed length scale.

Quadratic Lagrange elements are used for both fields, providing second-order accuracy with fewer degrees of freedom than linear elements [21]. To avoid numerical dispersion, a minimum of five elements per shear wavelength is maintained at the highest simulated frequency [22]. Where wave propagation occurs in a preferred direction, a structured mesh aligned with the wave fronts minimises numerical diffusion. Anisotropic scaling factors further allow resolution to be concentrated along the propagation direction. Model-specific mesh details are given in Section 4.

4 Materials and Methods

The methodology follows the structure illustrated in Figure 1: geometry selection, model configuration (physics, material, mesh), study design, and data extraction.

4.1 Geometry Selection

Three benchmark domains were implemented to isolate distinct contributions to the simulated wave field. The two-dimensional square (25×25 mm) serves as the baseline configuration: its regular, polygonal boundary makes it the simplest case for evaluating PML and periodic boundary-condition performance, and

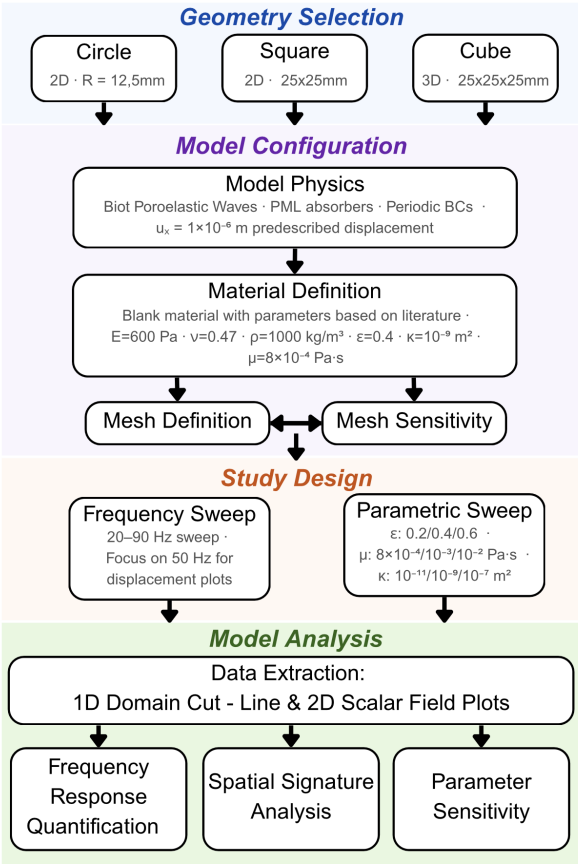


Figure 1. *Overview of the modelling workflow, from geometry selection through to the three analysis streams.*

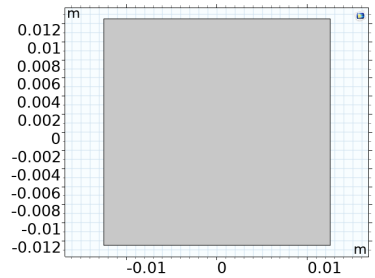


Figure 2. *Two-dimensional square domain (25×25 mm) as implemented in COMSOL.*

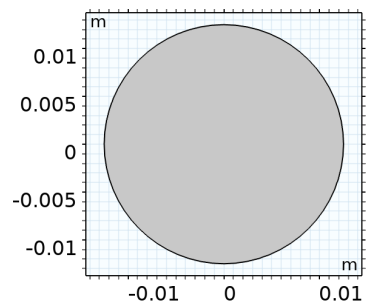


Figure 3. *Two-dimensional circular domain (diameter 25 mm) as implemented in COMSOL.*

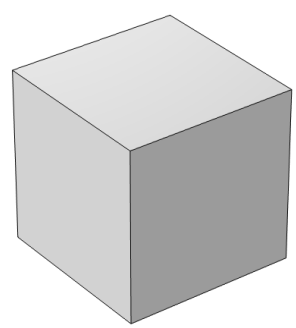


Figure 4. *Three-dimensional cube domain (side length 25 mm) as implemented in COMSOL.*

its structured mesh provides a direct reference for the numerical resolution study. The two-dimensional circle (diameter 25 mm, radius $R = 12.5$ mm) is introduced to isolate the effect of domain geometry while keeping the overall length scale consistent: its smooth, curved boundary and absence of corners make it a simplified proxy for a brain cross-section, and the direct square-versus-circle comparison in 2D isolates curvature effects from any change in material behaviour or frequency range. The three-dimensional cube ($25 \times 25 \times 25$ mm) extends the study to 3D at the same characteristic length, enabling the square-versus-cube comparison to isolate the effect of dimensionality under otherwise identical conditions. The three geometries are shown in Figures 2–4.

4.2 Model Physics

All three models are implemented in COMSOL Multiphysics using the Poroelastic Waves interface (Acoustics → Elastic Waves), which solves the coupled $(\mathbf{u}, p_f)$ Biot system in the frequency domain (Section 3). A parametric frequency sweep from 10 Hz to 100 Hz is performed, with each step solved as an independent steady-state harmonic problem. Quadratic Lagrange elements are used for both fields. The Biot viscous losses model is selected rather than the Biot–Allard alternative, the latter being intended for gas-saturated media. The viscosity model is set to the Biot low frequency range, appropriate for the 10–100 Hz regime where viscous forces in the pores dominate over inertial fluid effects.

Excitation and general boundary conditions

In all models the wave is introduced by prescribing a harmonic displacement $u_x = 1 \times 10^{-6}\,\mathrm{m}$ on one boundary, consistent with micron-scale displacements measured in clinical MRE. Zero transverse displacement is applied on faces perpendicular to the propagation direction to suppress rigid-body motion. A zero pore-pressure condition $p_f = 0\,\mathrm{Pa}$ is applied on all open (non-PML, non-periodic) boundaries, representing a drained surface.

PML configuration

PMLs are applied to attenuate outgoing waves at the domain boundaries in the propagation direction, preventing reflections that would otherwise contaminate the interior wave field with standing-wave artefacts. Absorbing the wave energy before it reflects from the boundary is essential at the top (excitation) and bottom faces, where the shear wave enters and exits the domain respectively; without damping on these faces, reflected energy would interfere destructively or constructively with the propagating front, making the interior field uninterpretable.

In COMSOL's polynomial coordinate stretching formulation, two parameters govern PML performance: the scaling factor and the curvature parameter. The scaling factor controls the overall amplitude of the complex coordinate stretch; the curvature parameter controls how steeply the stretching function grows across the PML thickness. A steeper profile increases the absorption gradient and reduces residual reflections at the discretised PML interface, which is particularly important in soft, highly attenuating media such as brain tissue where the shear wave amplitude already decays rapidly. In stiffer media, the wave amplitude decays more slowly over the same physical thickness, so a lower curvature can be sufficient to produce the required attenuation; conversely, in a very soft medium the steep curvature compensates for the shorter effective decay length available within the PML region.

A curvature parameter of 3 and a scaling factor of 1.5 are used in all models, following COMSOL's recommendation that values between 3 and 5 improve low-frequency performance when mixed propagating and evanescent wave components are present. These values were verified to eliminate residual interface reflections visible at lower curvature settings during the iterative boundary-condition development described below.

For the square and cube, PMLs are applied as Cartesian layers on the top and bottom faces only. The four lateral faces carry periodic boundary conditions (set to continuity) that replicate the domain infinitely in the lateral directions, eliminating lateral reflections without an absorbing layer and approximating propagation through a laterally unbounded tissue slab. PML and periodic conditions cannot be applied to the same boundary simultaneously, as the absorbing layer requires outgoing waves while the periodic condition imposes a coupling between opposing faces; this constraint directly determines which faces receive each treatment. For the circle, where the curved boundary precludes a Cartesian PML, an annular PML of thickness 2.5 mm is placed concentrically around the domain, defined in COMSOL as a cylindrical PML with polynomial coordinate stretching.

The final boundary configurations in all three models were arrived at iteratively: intermediate wave fields were inspected visually to identify residual artefacts, and conditions were adjusted until the interior field was physically interpretable.

Table 1 summarises the boundary conditions applied across all three geometries. The fixed conditions — prescribed displacement, zero pore pressure on open boundaries, and transverse displacement constraints — were established at the outset and retained throughout. The iterative development focused exclusively on the absorbing and lateral boundary treatment, with PML and periodic conditions added in the final configuration to suppress the standing-wave artefacts present in the initial setup.

Table 1. *Boundary condition configurations for all three geometries. Fixed conditions (top) are identical across initial and final configurations. Modified conditions (bottom) show what was added in the final configuration to suppress boundary artefacts.*

Fixed conditions — unchanged throughout		
Geometry	**Boundary**	**Condition**
All	Excitation	$u_x = 1 \times 10^{-6}\,\mathrm{m}$, $Q = 0$
All	Open BCs	$p_f = 0\,\mathrm{Pa}$
Square & Cube	Bottom	$u_y = 0$
Circle	Support point	$u_y = 0$ at $y_{\min}$
Cube	Excitation	$u_y = u_z = 0$
Modified conditions — added in final configuration		
Geometry	**Boundary**	**Added in final**
2D Square	Bottom	Cartesian PML
	Sides	Periodic (continuity)
2D Circle	Outer boundary	Annular PML (cylindrical)
3D Cube	Bottom	Cartesian PML
	Sides	Periodic (continuity)

All PML regions: curvature = 3, scaling = 1.5.

4.3 Brain Tissue Material Model

A single homogeneous poroelastic material representative of brain tissue is assigned to all domains. The material parameters were selected based on stiffness values reported by Budday et al., whose measurements place brain-tissue shear moduli in the range of hundreds of Pascals. The parameters are listed in Table 2. The near-incompressible Poisson ratio ($\nu = 0.47$) reflects the high water content of brain tissue. The Biot–Willis coefficient $\alpha_B = 1$ is consistent with the soft-tissue limit (Section 3). The tortuosity $\tau = 2$ accounts for the tortuous path fluid must travel through the pore network.

Table 2. *Material parameters representing brain tissue.*

Parameter	Symbol	Value
Young's modulus	E	600 Pa
Poisson ratio	ν	0.47
Solid density	ρ_s	$1000\,\mathrm{kg\,m^{-3}}$
Fluid density	ρ_f	$1000\,\mathrm{kg\,m^{-3}}$
Biot–Willis coefficient	α_B	1
Porosity	ε_p	0.4
Fluid compressibility	χ_f	$1 \times 10^{-5}\,\mathrm{Pa^{-1}}$
Permeability	κ	$1 \times 10^{-9}\,\mathrm{m^2}$
Fluid viscosity	η_f	$8 \times 10^{-4}\,\mathrm{Pa\,s}$
Tortuosity	τ	2

4.4 Mesh Definition and Sensitivity

Resolution criterion

The minimum mesh resolution required to avoid spatial aliasing is determined by the Nyquist–Shannon criterion applied to the shortest shear wavelength in the sweep. The shear wavelength at frequency f is

$$\lambda_s(f) = \frac{c_s}{f}, \qquad c_s = \sqrt{\frac{G_d}{\rho_{\mathrm{av}}}}, \tag{7}$$

where $G_d = E/[2(1+\nu)]$ is the drained shear modulus and $\rho_{\mathrm{av}} = (1-\varepsilon_p)\rho_s + \varepsilon_p\rho_f$ is the average saturated density. With the baseline parameters this gives $c_s \approx 0.45\,\mathrm{m\,s^{-1}}$ and a minimum wavelength $\lambda_s(90\,\mathrm{Hz}) \approx 4.9\,\mathrm{mm}$ at the highest simulated frequency. The number of elements per wavelength is therefore

$$n_{\mathrm{el}} = \frac{\lambda_s(f)}{\Delta x}, \tag{8}$$

where Δx is the characteristic element size. Numerical stability and prevention of spatial aliasing require a minimum of approximately 6 elements per wavelength for quadratic Lagrange elements [22]; 5 is the hard lower bound and values of 6–10 are the recommended working range. The target element size of 0.625 mm (40 elements over 25 mm) gives $n_{\mathrm{el}} \approx 7.9$ at 90 Hz, satisfying this criterion across the full sweep.

Circle mesh

A Free Triangular mesh is used for the circular interior domain, with the global element size set to *Finer* to limit numerical dispersion over the frequency sweep. Along the outer boundary, a boundary-layer mesh is added to increase near-boundary resolution, producing predominantly quadrilateral elements close to the perimeter with tight spacing. Ten layers with a stretching factor of 1.1 and a total thickness of 10 mm are used to capture steep displacement and pore-pressure gradients near the boundary while keeping the interior mesh computationally tractable. The mesh transitions from the quadrilateral boundary region into the triangular interior, providing refinement where gradients are largest without enforcing a uniformly fine discretisation.

Circle mesh sensitivity study. The critical numerical constraint in the circular geometry is the resolution of the elements at the interface between the circular domain and the PML annulus. Because the interior is discretised with an unstructured triangular mesh, its element size and orientation at the outer boundary cannot be controlled directly. An element-size mismatch at this transition introduces spurious reflections that contaminate the interior wave field and are indistinguishable from genuine boundary-driven attenuation.

Two alternative discretisation strategies for the PML annulus were compared against the boundary-layer approach to identify the most effective configuration:

- **Free Triangular mesh**: An unstructured triangular mesh applied uniformly across the circular interior.
- **Free Quad mesh** (global size: *Extremely Fine*): Quadrilateral elements are used in the PML annulus. However, the element size at the circular–PML interface is inherited directly from the interior triangular mesh, so that achieving adequate interface resolution requires refining the entire circular interior to *Extremely Fine*. This incurs a disproportionate computational cost throughout the domain, including regions far from the interface where such resolution is not required.

- **Boundary-layer mesh** (global size: *Finer*): Structured quadrilateral elements are generated near the outer boundary using a boundary-layer mesh, while the interior retains the coarser free triangular discretisation. The boundary-layer mesh decouples the interface resolution from the interior element size, concentrating high-resolution quadrilateral elements exclusively at the circular–PML transition. This yields equivalent suppression of interface artefacts at substantially lower computational cost than the *Extremely Fine* quad approach.

The boundary-layer mesh at *Finer* interior resolution was therefore selected as the working configuration for all circle simulations. The PML annulus is discretised with a structured mesh, with resolution increasing towards the inner PML boundary to reduce spurious reflections and improve low-frequency performance. The resulting mesh is shown in Figure 5.

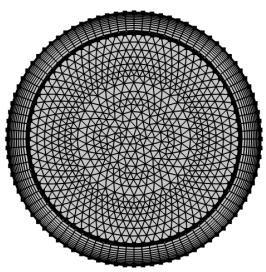


Figure 5. *Mesh of the circular domain including the annular PML.*

Square and cube meshes

The square and cube share the same structured meshing approach, both targeting the 0.625 mm element size derived from the criterion above. The square domain is discretised using a Mapped Mesh of 40×40 elements over the 25×25 mm core, giving 7.9 elements per shear wavelength at 90 Hz. The same 40-element horizontal distribution is projected through the PML depth, with 10 elements in the vertical PML thickness, maintaining nodal continuity at the tissue–PML interface and eliminating impedance mismatches that would otherwise generate artificial reflections. The resulting mesh is shown in Figure 6.

The cube follows the same logic extended to three dimensions. A Mapped mesh is applied to the xz-face with distributions of 40 elements in x and 2 elements in z, then swept through the domain in the y-direction with 40 elements, producing a structured hexahedral mesh of $40 \times 40 \times 2$ elements. The PML layers are swept separately with 10 elements, keeping the interface treatment identical to the square. In the z-direction only 2 elements are used: since the periodic boundary conditions on both z-faces confine wave propagation to the in-plane directions, out-of-plane resolution is not a primary modelling concern. Increasing the z-resolution to 5 elements caused solver memory failures on the available hardware, making two elements both physically justified and the computational limit. The resulting mesh is shown in Figure 7.

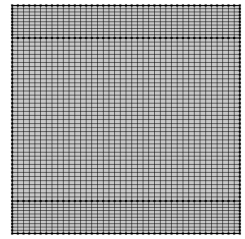


Figure 6. *Mesh of the square domain including PMLs.*

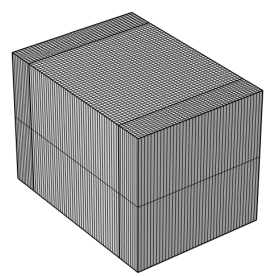


Figure 7. *Structured hexahedral mesh of the cube domain.*

Because the z-direction is specific to the 3D geometry and cannot be validated by the 2D criterion, a mesh sensitivity study was conducted on the cube, as described below.

Cube mesh sensitivity study

The mesh sensitivity study tested three configurations to verify that the selected $40 \times 40 \times 2$ mesh (x, y, z elements) satisfies the Nyquist spatial resolution criterion across the full frequency range, and that the choice of $z = 2$ layers does not introduce significant numerical error. The three configurations compared were:

- $40 \times 40 \times 2$ — baseline ($\approx 8\,\mathrm{el}/\lambda$ at 90 Hz in both x and y),
- $40 \times 20 \times 2$ — halved y-resolution ($\approx 4\,\mathrm{el}/\lambda$ in the propagation direction),

- $20 \times 40 \times 2$ — halved x-resolution ($\approx 4\,\mathrm{el}/\lambda$ in the shear-displacement direction).

The coarser alternatives are designed to test what happens when the criterion is violated in each of the two in-plane directions independently, while $z = 2$ is held fixed throughout as the computationally constrained value. Additionally, z-uniformity was assessed separately by extracting the displacement profile along a z-direction cutline at multiple frequencies, verifying that the field does not vary meaningfully across the two z-layers under the applied periodic conditions.

For the in-plane comparison, displacement magnitude was exported along a y-direction cutline through the domain centre at 20 Hz, 50 Hz and 90 Hz. Deviation from the baseline was quantified as the normalised absolute difference relative to the peak baseline displacement. The $40 \times 40 \times 2$ mesh was selected on the basis that it satisfies the criterion, remains computationally feasible, and produces results consistent with the coarser meshes at lower frequencies while diverging predictably at higher frequencies where the criterion begins to be violated. The outcomes of this comparison are reported in Section 5.2.3.

4.5 Study Design

Following mesh validation, two complementary analysis streams were carried out, corresponding to the two right-hand branches of the workflow in Figure 1.

The first stream is a **frequency sweep** performed on all three geometries from 20 Hz to 90 Hz in 1 Hz steps. This range covers the clinically relevant brain MRE excitation band and extends to 90 Hz where the shear wavelength becomes short enough that boundary treatment and geometric effects are most pronounced. Displacement magnitude plots at 50 Hz are used as the primary visual reference for spatial field quality, as this frequency places multiple wavelengths within the domain while remaining well within the clinical range. The frequency sweep enables the cross-geometry comparison that addresses research question 2.

The second stream is a **parametric sweep** conducted across all three geometries, varying three material parameters (porosity ε_p, fluid viscosity η_f, and hydraulic permeability κ) independently across three levels each, evaluated at four frequencies (20, 50, 70, and 90 Hz), giving $3 \times 3 \times 3 \times 4 = 108$ simulations per geometry. These three parameters were selected because they are the primary determinants offluid–solid coupling in Biot's framework (governing how interstitial fluid moves relative to the solid matrix, how much energy is dissipated, and how pressure builds up in the pore space) and because they carry the widest reported uncertainty ranges in the brain tissue literature, being sensitive to physiological state (e.g. oedema, sleep cycle, and disease). Young's modulus, Poisson ratio, and density were not included in the sweep because they are among the best-constrained brain tissue parameters reported in the literature, and their primary role in determining the baseline shear wave speed is already represented by the reference simulation using the fixed values given in Table 2. Running the parametric sweep across all three geometries allows material sensitivity to be assessed in conjunction with the geometric effects studied in the frequency-sweep stream.

The parameter levels were chosen to span the physiologically plausible range with steps large enough to reveal distinct physical regimes rather than small perturbations:

- ε_p: 0.2, 0.4, 0.6 — linear steps of ± 0.2 centred on the baseline, spanning from a relatively dense solid matrix to a nearly fluid-dominated tissue.
- η_f: 8×10^{-4}, 8×10^{-3}, $8 \times 10^{-2}\,\mathrm{Pa\,s}$ — decade steps from the CSF/water baseline upward, covering the range reported across different fluid compositions and pathological states.
- κ: 1×10^{-11}, 1×10^{-9}, $1 \times 10^{-7}\,\mathrm{m}^2$ — steps of $\times 100$ spanning the full reported brain tissue range, representing regimes from near-impermeable (fluid trapped, high wave speed) to highly permeable (free fluid flow, reduced coupling).

The scope of the sweep was also constrained by computational time, as each simulation requires solving the full coupled Biot system across the domain at a given frequency, and 108 runs already represents a significant batch. Varying additional parameters or finer steps would have been informative but was not feasible within the project timeline.

4.6 Data Extraction

All quantitative analysis is based on displacement data extracted along a one-dimensional cutline drawn through the interior of each domain parallel to the wave propagation direction (y-axis). The cutline passes through the domain centre and is deliberately kept away from both the PML regions and the excitation boundary to avoid contamination from boundary-imposed values: at the excitation face the prescribed displacement is a boundary condition, not a simulation result, and within the PML the field is

artificially attenuated by design. For all three geometries the cutline therefore runs from $y = -12.2\,\text{mm}$ to $y = +12.2\,\text{mm}$, leaving a 0.3 mm margin from the domain boundary at either end. For the square and cube this places the endpoints 0.3 mm inside the PML interface; for the circle it keeps the endpoints within the circular boundary away from the curved PML annulus. The cutline placement is illustrated in Figure 8.

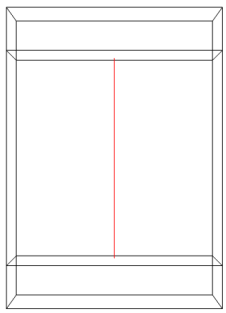


Figure 8. *Cutline placement illustrated on the 3D cube domain. The line runs along the y-axis through the domain centre from $y = -12.2\,\text{mm}$ to $y = +12.2\,\text{mm}$, avoiding both the excitation boundary and the PML regions. The same convention is applied to the square and circle.*

Three types of data are extracted from the cutline:

1. **Spatial displacement profile at 50 Hz.** Displacement magnitude $\|\mathbf{u}\|$ is plotted against arc length along the cutline at 50 Hz. This profile is used to assess wave field quality, wavelength, and the degree of attenuation across the domain for each geometry and boundary configuration.

2. **Frequency response.** The maximum displacement magnitude along the cutline is extracted at each frequency step of the 10–100 Hz sweep, yielding a single scalar value per frequency. Plotting this against frequency reveals the resonance structure of the domain and enables cross-geometry comparison of frequency-dependent behaviour.

3. **Parametric sweep response.** For the parametric simulations, the maximum cutline displacement is extracted at each of the four evaluation frequencies (20, 50, 70, 90 Hz) for every parameter combination across all three geometries. This dataset is the basis for the material parameter sensitivity analysis in Section 5.5.

In addition to the cutline data, two-dimensional (and three-dimensional mid-plane) displacement magnitude plots were exported directly from COMSOL at 50 Hz for each geometry and boundary configuration. These spatial field visualisations are used qualitatively to assess wave-front regularity and identify standing-wave artefacts. Together, the cutline and field data form the basis for the three analysis streams presented in the results: Frequency Response Quantification, Spatial Signature Analysis, and Parameter Sensitivity Analysis.

5 Results

5.1 Comparative Analysis Across Geometries

Figure 9 overlays the one-dimensional displacement magnitude profiles along the propagation-direction centreline at 50 Hz for all three geometries, each extracted under its respective final boundary configuration. The square and cube show almost identical

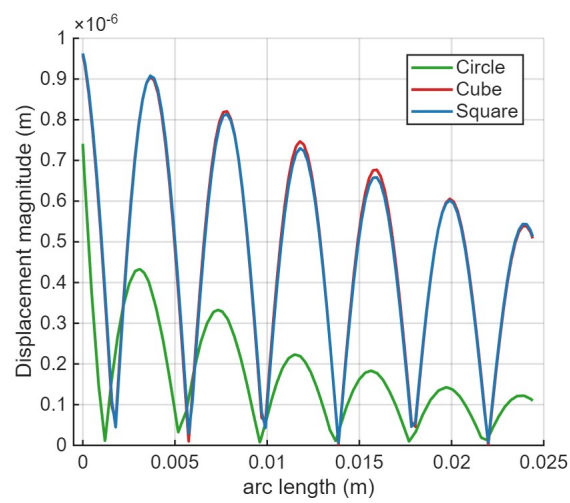


Figure 9. *Cutline displacement magnitude $\|\mathbf{u}\|$ at 50 Hz for all three geometries*

displacement profiles along the full 2.5 cm cutline. Peak positions, troughs, and the overall decay envelope coincide closely, with no visible difference in spatial wavelength or amplitude. By contrast, the circular domain exhibits a shorter spatial wavelength, lower displacement amplitudes, and a steeper decay along the propagation path, consistent with the reduced phase velocity reported in Table 3. As a result, the displacement magnitude at the end of the cutline is markedly lower in the circular domain than in the square and cube.

5.2 Geometry-Specific Results

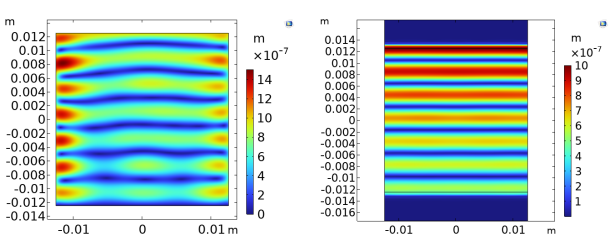
The cross-geometry comparison based on the final boundary configurations indicates similar displacement behaviour in the square and cube, while the circular domain shows a distinct response. The following subsections examine the boundary-condition development for each geometry and compare the initial setups with the final configurations.

5.2.1 Two-Dimensional Square

The boundary condition development for the square domain was evaluated by comparing the displacement magnitude $\|\mathbf{u}\|$ at 50 Hz. This frequency provides a rigorous test of mesh density and boundary efficiency, as multiple wavelengths are contained within the 2.5 cm domain.

In the initial configuration (Figure 10a), characterised by hard-wall boundaries, the wave field is dominated by a disordered interference pattern. Reflections from the lateral and bottom edges produce localised hotspots of amplified displacement and non-physical standing-wave nodes, obscuring the primary shear-wave front. The lack of nodal alignment at the domain interfaces triggers spurious scattering that masks the intrinsic poroelastic response of the tissue.

The final configuration, utilising the hybrid strategy of top/bottom PMLs and lateral periodic conditions, produces a significantly more coherent field (Figure 10b). Wave fronts appear as regular parallel horizontal bands propagating vertically through the domain. The transition into the PML regions is smooth, confirming that the 40×40 mapped mesh and nodal alignment at the tissue–PML interface successfully eliminate interface reflections. The spatially periodic fronts demonstrate that the domain effectively simulates a laterally unbounded tissue section, free from the constraints of a finite box geometry.



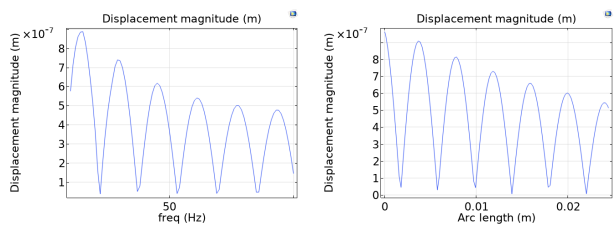
(a) *Initial: strong reflection patterns dominate the interior field.* **(b)** *Final: smoother wave propagation and strongly reduced boundary artefacts.*

Figure 10. *Displacement magnitude* $\|\mathbf{u}\|$ *at* 50 Hz *for the 2D square model.*

To evaluate the model's performance beyond the 2D surface field, quantitative metrics were extracted from the domain centre $(0, 0)$ through a frequency sweep and a centreline spatial profile.

Figure 11a displays the displacement magnitude at the central cut point as a function of frequency from 1 to 100 Hz. A primary peak is observed at approximately 18 Hz, identifying the fundamental resonance of the domain. At higher frequencies, the curve exhibits characteristic comb-like spikes with sharp nulls occurring approximately every 15 Hz. Because the measurement is taken at a single fixed point, these nulls indicate frequencies at which a standing-wave node falls exactly at the domain centre. The uniform spacing of the peaks corresponds to the harmonic modes of the 2.5 cm geometry, and the same resonance structure is observed in the cube (Section 5.2.3).

Figure 11b presents the displacement magnitude along a vertical cutline at 50 Hz. The profile reveals a decaying oscillatory envelope as the distance from the excitation edge increases, illustrating the intrinsic attenuation within the poroelastic matrix. The magnitude effectively drops to zero as the wave enters the PML, confirming the layer's ability to absorb outgoing energy without creating return reflections.



(a) *Frequency sweep at central cut point* $(0, 0)$. **(b)** *Spatial 1D cutline at* 50 Hz.

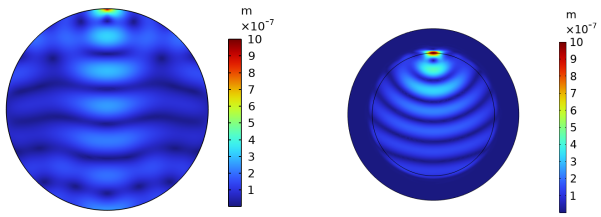
Figure 11. *Quantitative evaluation of the final square model configuration.*

5.2.2 Two-Dimensional Circle

The boundary condition development for the circular domain was evaluated by comparing the displacement magnitude $\|\mathbf{u}\|$ at 50 Hz. Unlike the square, the smooth curved boundary continuously redirects reflections toward the interior from all directions simultaneously, making the initial interference pattern more severe and preventing the use of periodic lateral conditions.

In the initial configuration (Figure 12a), the interior response is dominated by spatially irregular interference features. The field contains multiple superimposed components that produce locally amplified zones and strongly distorted wave fronts across the entire disk. In the final configuration (Figure 12b),

the interior pattern becomes markedly more coherent: wave fronts are smoother and the visually apparent standing-wave content is reduced, yielding a cleaner field more suitable for subsequent wavelength-based interpretation. Compared with the square final field, the fronts are less uniform and the amplitude visibly lower across the domain, consistent with the cross-geometry comparison in Figure 9.



(a) *Initial: strong interference patterns dominate the interior field.*

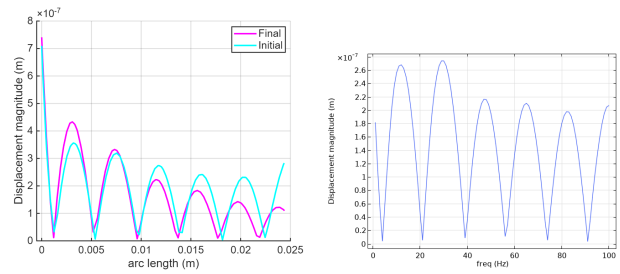
(b) *Final: smoother wave fronts and reduced boundary artefacts.*

Figure 12. *Displacement magnitude* $\|\mathbf{u}\|$ *at* 50 Hz *for the 2D circle model.*

To evaluate the model's performance beyond the 2D surface field, quantitative metrics were extracted from the centreline ($x = 0$) through a frequency sweep and a centreline spatial profile.

Figure 13b displays the displacement magnitude at the centreline measurement point as a function of frequency. The curve exhibits a periodic oscillation with peaks occurring approximately every 15–17 Hz, consistent with the harmonic mode spacing observed in the square and cube. However, in contrast to the sharp comb-like spikes and deep nulls of those geometries, the circle response is considerably smoother: the nulls do not reach zero and the envelope remains broadly flat across the sweep. This reflects the absence of periodic lateral conditions — without them, energy is not fully confined to a standing-wave pattern between opposing faces, so destructive interference at the measurement point is incomplete and the resonance peaks are damped rather than sharp.

Figure 13a shows the centreline displacement magnitude at 50 Hz for both configurations overlaid. Both profiles share the same spatial wavelength and near-identical peak positions, confirming that the underlying shear wave speed is unchanged between configurations. The final configuration exhibits a consistently lower envelope and steeper decay than the initial, indicating that the annular PML successfully reduces the reflected energy recirculating within the domain. In contrast to the square cutline, the envelope decays more steeply and the peak amplitudes are lower throughout — a difference that persists in the final configuration and is consistent with the absence of periodic lateral boundary conditions.



(a) *Centreline* $\|\mathbf{u}\|$ *at* 50 Hz*, initial and final configurations overlaid.*

(b) *Frequency sweep at the centreline measurement point.*

Figure 13. *Quantitative evaluation of the final circle model configuration.*

5.2.3 Three-Dimensional Cube

Mesh sensitivity. The mesh sensitivity study compared three configurations — $40 \times 40 \times 2$, $40 \times 20 \times 2$ (halved y-resolution), and $20 \times 40 \times 2$ (halved x-resolution) — as described in Section 4.4. Mesh sensitivity effects are inherently frequency-dependent, as frequency increases the shear wavelength shortens, placing progressively greater demand on the spatial resolution, and any under-resolved configuration will accumulate phase and amplitude error more rapidly. The 90 Hz case therefore represents the most stringent test and is the primary basis for the mesh selection and result analysis.

Figure 14 shows the normalised deviation from the $40 \times 40 \times 2$ mesh at 90 Hz for both coarser configurations. The $40 \times 20 \times 2$ mesh, which reduces the propagation-direction resolution to $4\,\mathrm{el}/\lambda$, reaches deviations that repeatedly exceed 5%, consistent with Nyquist failure in the direction the wave travels. The $20 \times 40 \times 2$ mesh, which halves the resolution in the shear-displacement direction instead, remains substantially better behaved, confirming that it is specifically the propagation-direction resolution that governs accuracy at high frequency.

The z-direction uniformity was assessed by extracting the displacement profile along a z-direction cutline at 20, 50, and 90 Hz (Annex, Figure 19). Uniformity was quantified as the normalised range across the two z-layers,

$$\delta_z = \frac{\max\|\mathbf{u}\|_z - \min\|\mathbf{u}\|_z}{\max\|\mathbf{u}\|_y}, \tag{9}$$

where $\max\|\mathbf{u}\|_z$ and $\min\|\mathbf{u}\|_z$ are the maximum and minimum displacement magnitudes along the z-

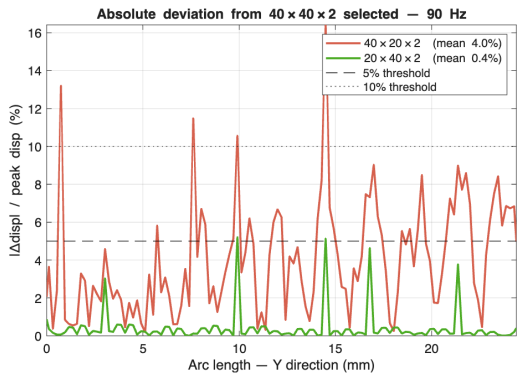
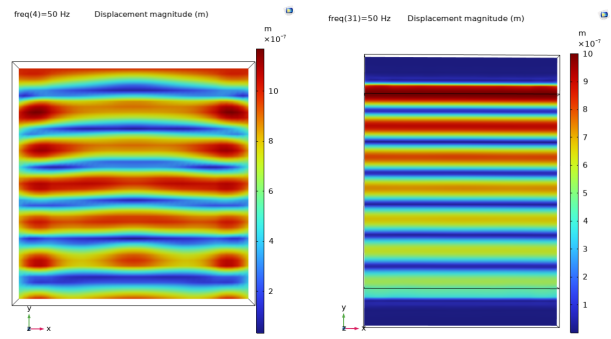


Figure 14. *Normalised deviation from the* $40 \times 40 \times 2$ *mesh at* 90 Hz *for the two coarser configurations, computed as the absolute difference relative to the peak displacement. The* $40 \times 20 \times 2$ *configuration repeatedly exceeds* 5%*, indicating Nyquist failure in the propagation direction.*

cutline, and $\max\|\mathbf{u}\|_y$ is the peak displacement along the y-direction cutline at the same frequency, used as the reference scale. This normalisation ensures the variation is expressed relative to the physically meaningful wave amplitude rather than the near-zero out-of-plane field. δ_z remains below 0.5% at 20 and 50 Hz. At 90 Hz it reaches approximately 9%; however, $\max\|\mathbf{u}\|_z \approx 0.025\,\mu$m at this frequency compared to $\max\|\mathbf{u}\|_y \approx 0.9\,\mu$m, so the numerator of Equation 9 is itself a near-zero quantity. The apparent 9% therefore reflects numerical noise on an essentially flat out-of-plane field rather than a physically meaningful variation. Together, the in-plane and z-uniformity results confirm that the $40 \times 40 \times 2$ mesh satisfies the resolution requirements across the full frequency sweep without introducing out-of-plane error.

Wave field. The boundary condition development for the cube was evaluated using mid-plane cross-sections at 50 Hz. Starting from only the displacement excitation and zero pressure condition, the field is dominated by standing-wave interference from reflections at all six faces. Introducing top and bottom PML with lateral periodic conditions produces a visually regular wave field with spatially periodic fronts and smooth amplitude variation throughout the domain. The contrast between the two configurations is shown in Figure 15.

Figure 16 shows the displacement magnitude along the cutline at 50 Hz for the final configuration, confirming a smooth decaying wave profile through the interior of the domain with no boundary-driven distortion.



(a) *Initial: standing-wave interference.* **(b)** *Final: regular wave fronts.*

Figure 15. *Mid-plane cross-sections at* 50 Hz*, initial (left) and final (right) boundary configurations.*

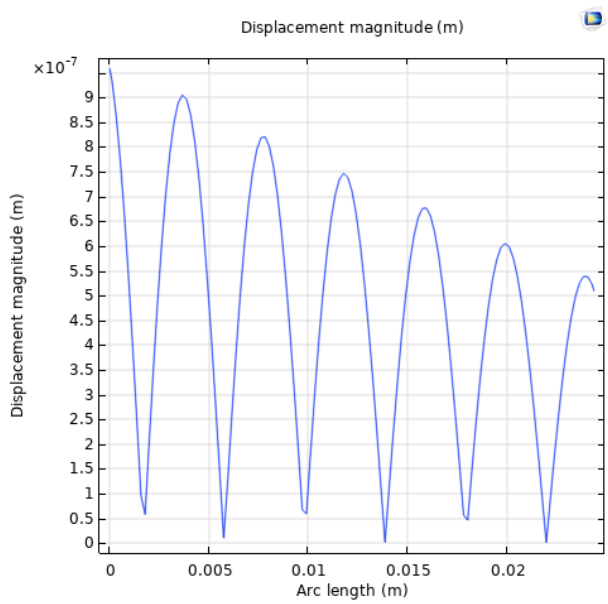


Figure 16. *Displacement magnitude* $\|\mathbf{u}\|$ *along the* y*-axis cutline at* 50 Hz*.*

5.3 Pillar A: Frequency Response Quantification

In this pillar, the three benchmark geometries are evaluated by quantifying their resonant characteristics and damping characteristics. While the initial frequency sweeps identified discrete peaks in displacement, this analysis provides the physical valuation of those responses.

The primary metric for frequency response is the Amplification Factor (A), defined as the ratio of the maximum steady-state displacement magnitude ($|u_{max}|$) extracted along the domain cutline to the prescribed

boundary input ($u_{in} = 1 \times 10^{-6}$ m):

$$A = \frac{|u_{max}|}{u_{in}}, \tag{10}$$

The 2D Square was the only geometry to demonstrate a value of $A > 1$ at the reference frequency (f_0) of 50 Hz, reaching a peak displacement of 1.00077×10^{-6} m. Its fundamental resonance was identified at approximately 14 Hz (Table 3). The 3D Cube exhibited a peak displacement of 0.99759×10^{-6} m at 44 Hz, while the 2D Circle showed the highest level of signal attenuation, with a peak of 0.75147×10^{-6} m at 41 Hz.

Table 3. *Frequency Response: Key Metrics.*

Geometry	$f_{\mathbf{res}}$ (Hz)	$u_{\mathbf{max}}$ (10^{-7} m)	A	**FWHM** (Hz)
2D Square	14	10.0	1.001	50
2D Circle	11	7.52	0.751	41
3D Cube	16	9.98	0.998	44

† *2D Square: resonant; others: damped.*

5.3.1 Peak Spread & System Stability

System stability was quantified using the Full Width at Half Maximum (FWHM), or "Peak Spread", which measures the frequency bandwidth over which the displacement remains above a calculated threshold. The threshold was defined as the midpoint between the peak magnitude and the noise floor ($|u_{min}|$):

$$\text{Threshold} = \frac{|u_{max}| + |u_{min}|}{2}, \tag{11}$$

$$\Delta f = f_{upper} - f_{lower} \quad \text{(FWHM)} \tag{12}$$

The 2D Square displayed the widest frequency spread at its 50 Hz peak, while the 3D Cube (44 Hz) demonstrated an intermediate spread. Finally, the 2D Circle (41 Hz) yielded the narrowest spread.

5.3.2 Damping Hierarchy & Quality Factor (Q)

In order to quantify energy dissipation, the Quality Factor (Q) was calculated using the 3dB bandwidth method:

$$Q = \frac{f_0}{\Delta f} \tag{13}$$

The analysis confirmed a consistent damping hierarchy across all geometries, with calculated Q values remaining well below unity.

- 2D Square: $Q = 0.278$;
- 3D Cube: $Q = 0.286$;
- 2D Circle: $Q = 0.333$.

5.4 Pillar B: Spatial Signature Analysis

Pillar B evaluates the spatial propagation of the shear wave across the three benchmark geometries at a baseline frequency of 50 Hz. The analysis quantifies the rate of energy dissipation as the wave travels from the excitation boundary through the poroelastic medium.

5.4.1 Quantification of Wave Attenuation

To isolate the intrinsic material damping from the complex interference patterns observed in the raw nodal data, a multi-step data reduction process is employed. First, a peak-detection algorithm extracts the wave crests along the domain cutline to define the decay envelope of the signal. To ensure numerical accuracy, a spatial "filter" is applied to the first 3 mm of the domain to exclude near-field artifacts (localised pressure spikes occurring at the tissue-excitation interface). The filtered "Far-Field" peaks are then fitted using a non-linear least squares regression based on the exponential attenuation law:

$$|u(y)| = A \cdot e^{-\alpha(y-y_0)} \tag{14}$$

where $|u(y)|$ is the displacement magnitude at depth y, A is the fitted amplitude at the start of the far-field region (y_0), and α is the Attenuation Coefficient (m^{-1}). From this primary regression, two secondary physical constants are derived to characterize the spatial signature:

- **Penetration Depth (δ)**: The distance required for the signal magnitude to drop to $\approx 37\%$ ($1/e$) of its original value, calculated as $\delta = 1/\alpha$.
- **Phase Velocity (c_p)**: Extracted by measuring the mean spatial distance between consecutive peaks (Wavelength, λ) at the 50 Hz excitation frequency:

$$c_p = f \cdot \lambda \tag{15}$$

5.4.2 Spatial Metrics Results (at 50 Hz)

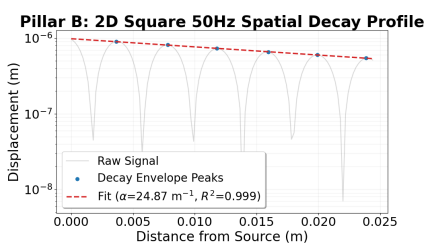
The extraction of these spatial metrics across the final model configurations, focusing on our reference frequency of 50 Hz, yielded the results shown in Table 4.

The results demonstrate a high degree of quantitative convergence between the rectilinear models. The

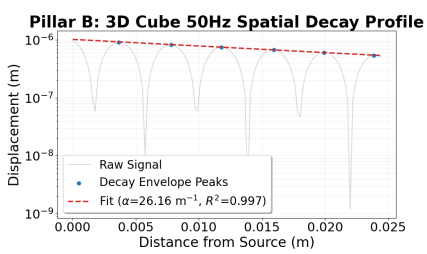
Table 4. *Wave propagation properties per model geometry.*

Geometry	α (m^{-1})	δ (mm)	c_p (m/s)
2D Square	24.87	40.22	≈ 2.01
3D Cube	26.16	38.23	≈ 1.91
2D Circle	48.05	20.81	≈ 0.44

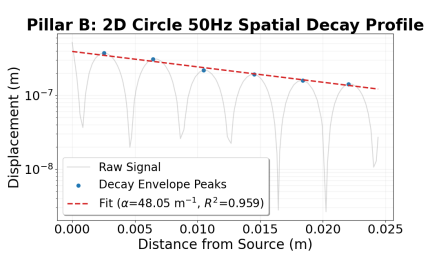
attenuation coefficients for the 2D Square and 3D Cube differ by only 5%, validating the 2D plane-strain model as a computationally efficient surrogate for capturing volumetric decay. In contrast, the circular geometry exhibits a significantly higher attenuation coefficient ($\alpha = 48.05$ m^{-1}), nearly doubling the dissipation rate observed in the flat-boundary models. This results in a penetration depth of 20.81 mm, representing a 50% reduction in wave reach compared to the Square. Notably, while the wave in rectilinear geometries reaches the distal boundary (25 mm) with substantial energy ($\delta \approx 40$ mm), the wave in the circular domain decays almost entirely within the physical limits of the tissue.



(a) *2D Square Decay Profile*



(b) *3D Cube Decay Profile*



(c) *2D Circle Decay Profile*

Figure 17. *Geometries Decay Profiles*

5.5 Pillar C: Material Parameter Sensitivity

In this final Pillar, we evaluate the mechanical robustness of each geometry by assessing variations in the primary poroelastic parameters (Fluid Viscosity η_f, Porosity ϵ_p, and Permeability κ) and their effect on the displacement field. This analysis utilised a 108-scenario batch simulation per geometry to define the sensitivity indices across the clinical frequency range (20, 50, 70, and 90 Hz).

Two primary statistical metrics were employed to quantify model behavior:

- Sensitivity Index ($S\%$): Measures the specific impact of an individual parameter on the mean displacement:

$$S = \frac{\mu_{max} - \mu_{min}}{\mu_{global}} \times 100$$

  where μ_{max} and μ_{min} are the maximum and minimum mean displacements found within the three levels of a single parameter.

- Coefficient of Variation ($CV\%$): Quantifies the global "Model Uncertainty" by expressing the standard deviation as a percentage of the global mean displacement for a given frequency:

$$CV\% = \frac{\sigma_{global}}{\mu_{global}} \times 100$$

5.5.1 Parametric Sweep

The execution of the parametric sweep revealed that Permeability (κ) is the co-dominant determinant of displacement magnitude across all geometries.

Table 5. *Sensitivity analysis and model uncertainty per geometry (at reference frequency* 50 Hz*).*

Geometry	Master Parameter	S (%)	CV (%)
2D Square	Permeability (κ)	4.57	3.17
3D Cube	Permeability (κ)	4.08	3.05
2D Circle	Permeability (κ)	4.73	2.66

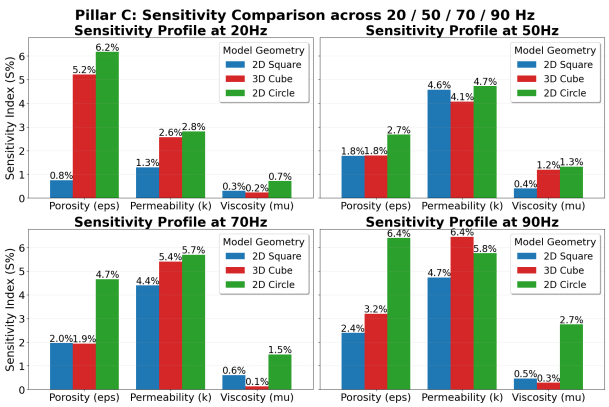
The parametric data identified three critical trends in the tissue's mechanical response:

- Permeability (κ) maintains the highest $S\%$ index across all shapes, identifying it as the primary controller of fluid-solid coupling in the poroelastic matrix.

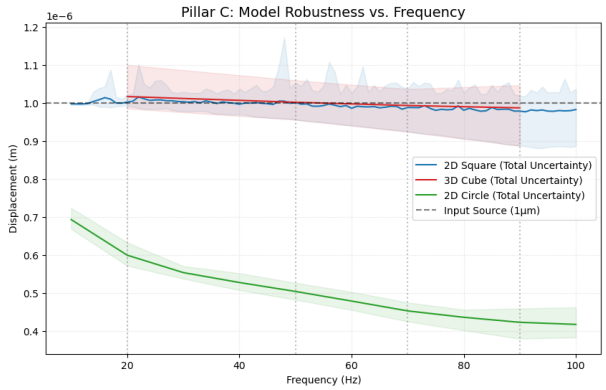
- Sensitivity to Porosity (ϵ_p) was observed to be higher at 20 Hz (reaching 6.2% in the Circle)

than at 90 Hz. This suggests that low-frequency oscillations allow significantly more time for interstitial fluid flow, heightening the impact of pore volume on damping.

- The global uncertainty ($CV\%$) remains remarkably stable across models, ranging from 2.66% to 3.17% at 50 Hz. However, the mean difference in sensitivity index between the 2D Square and 3D Cube across all parameter and frequency combinations amounts to 29.2 %. This indicates that while 2D models successfully predict global damping signatures, they diverge when quantifying specific material-parameter interactions.



(a) *Sensitivity comparison across 20, 50, 70 and 90 Hz for each geometry.*



(b) *Model robustness vs. frequency: uncertainty envelopes for each geometry.*

Figure 18. *Pillar C: sensitivity and robustness analysis.*

6 Discussion

The present study investigated shear-wave propagation in poroelastic brain tissue models using three benchmark geometries: a two-dimensional square, a two-dimensional circle, and a three-dimensional cube. By analysing frequency response, spatial attenuation, and parameter sensitivity, the simulations provide insight into how geometric configuration and material properties influence wave behaviour in computational brain tissue models.

Across the three analysis pillars, two key patterns emerge. First, rectilinear geometries (square and cube) produce highly consistent propagation behaviour, while the circular geometry introduces stronger attenuation and reduced displacement amplitudes. Second, permeability consistently emerges as the dominant parameter controlling displacement magnitude and energy dissipation within the poroelastic matrix.

The quantitative values obtained in the present simulations are broadly consistent with experimentally reported mechanical behaviour of brain tissue. In vivo magnetic resonance elastography (MRE) studies typically report shear wave speeds in the human brain on the order of approximately 1–2 m/s at excitation frequencies near 50 Hz, corresponding to shear moduli in the kilopascal range [23]. The phase velocities obtained for the rectilinear geometries in the present simulations (approximately 1.9–2.0 m/s) fall within this experimentally observed range, suggesting that the selected material parameters and numerical configuration produce physically realistic wave propagation behaviour.

6.1 Impact of Geometry: Scattering vs Resonance

One of the clearest observations from the results is the strong influence of domain geometry on the propagation and attenuation of shear waves. The square and cube geometries show almost identical displacement profiles along the propagation cutline, both exhibiting similar wavelengths, peak positions, and decay envelopes (Figures 17a, 17b and 18b).

This similarity indicates that the essential wave dynamics are governed primarily by the material properties rather than dimensionality when the boundary structure is rectilinear. The periodic boundary conditions applied laterally in these geometries effectively emulate an infinitely extended medium, allowing shear waves to propagate without significant lateral scattering.

In contrast, the circular domain produces a noticeably different response. Although the spatial wavelength within the circular domain remains unchanged between the initial and final boundary configurations, confirming that the boundary treatment does not alter the intrinsic shear wave speed, the displacement

amplitudes are consistently lower and decay more rapidly along the propagation direction than in the rectilinear geometries (see Table 4).

This behaviour can be attributed to geometric scattering. The curved boundary continuously redirects wave energy toward the interior of the domain, producing a more complex interference pattern and preventing the formation of clean standing-wave modes. Without periodic lateral constraints, energy is redistributed rather than trapped between parallel boundaries, leading to smoother frequency responses and reduced resonance peaks.

These results highlight an important modelling implication: domain geometry can significantly influence observed attenuation behaviour, even when the underlying material model remains identical. Consequently, circular domains may artificially enhance apparent damping due to geometric scattering rather than intrinsic tissue properties.

6.2 Dominance of Permeability in Poroelastic Damping

The parametric sensitivity analysis identifies permeability as the dominant parameter governing displacement magnitude across all geometries. The sensitivity indices consistently show the largest response to permeability variations, particularly at higher excitation frequencies.

This result aligns with the theoretical framework of Biot's poroelasticity, in which permeability controls the ease with which interstitial fluid moves through the porous solid matrix [16, 17]. When permeability increases, fluid pressure gradients dissipate more rapidly, reducing viscous drag between the fluid and solid phases. This directly alters the energy dissipation rate of the propagating shear wave.

The frequency dependence observed in the sensitivity analysis further supports this interpretation (Table 5, Figure 18a). At lower frequencies, porosity plays a slightly larger role because the oscillation period allows sufficient time for fluid redistribution within the pore space. As frequency increases, however, the fluid cannot fully equilibrate during each cycle, and the resistance to flow, governed by permeability, becomes the dominant damping mechanism. Interestingly, while permeability remains globally dominant, the circular geometry exhibits localized frequency-dependent sensitivity to porosity, particularly at 70 Hz and 90 Hz. This behaviour suggests that at specific excitation frequencies the interaction between the curved boundary and the internal pore pressure field may alter the balance between fluid storage and fluid flow mechanisms. In such conditions, the geometric confinement of the domain can lead to partial trapping of pore pressure oscillations, temporarily increasing the influence of porosity on the displacement response. This effect highlights that non-rectilinear geometries may introduce additional coupling between material parameters and wave dynamics, producing localized deviations from the permeability-dominated behaviour predicted.

The attenuation coefficients obtained from the spatial decay analysis fall within the range expected for highly damped biological soft tissues. Brain tissue exhibits strong viscoelastic and poroelastic dissipation due to the combined effects of matrix deformation, internal friction, and fluid flow through the extracellular space. Experimental elastography studies have similarly reported rapid amplitude decay of shear waves over relatively short propagation distances in soft tissues [4, 10].

The substantially shorter penetration depth in the circular domain (Table 4) cannot be attributed to differences in material dissipation alone, since identical parameters govern all three models. The additional attenuation arises from geometric energy redistribution: as the curved boundary continuously scatters wave energy inward, part of the propagating energy is permanently redirected away from the centreline measurement path, producing an apparent attenuation that exceeds the intrinsic poroelastic value.

The damping behaviour of the simulated systems is further reflected in the calculated quality factors obtained from the frequency-response analysis. The relatively low Q values observed across all geometries indicate that the system operates in a strongly damped regime, where viscous dissipation within the poroelastic matrix suppresses the development of high-amplitude resonance. In fluid-saturated biological tissues, energy loss occurs through two primary mechanisms: internal viscoelastic friction within the solid matrix and viscous flow of interstitial fluid through the pore network. These processes dissipate mechanical energy during each oscillation cycle, resulting in broader resonance peaks and reduced amplitude amplification compared with purely elastic materials [6, 10].

The close agreement between the Q values of the square ($Q = 0.278$) and cube ($Q = 0.286$) further supports the interpretation that damping behaviour is governed primarily by intrinsic material properties rather than dimensionality. By contrast, the circular geometry exhibits a notably higher Q value ($Q = 0.333$). This suggests that while the curved boundary increases spatial attenuation along the cen-

terline via scattering, it simultaneously acts to concentrate or "trap" wave energy through boundary-driven reflections. These reflections likely interfere constructively to sustain better-defined resonance peaks than those observed in rectilinear domains. Consequently, the circular geometry appears less damped in the frequency domain, even though its spatial displacement profile exhibits a steeper decay envelope due to geometric energy redistribution. The relatively small global coefficients of variation (approximately 2.6–3.2%, Table 5) indicate that the model response remains stable across the tested parameter ranges. This suggests that the numerical implementation is robust and that the predicted wave behaviour is not excessively sensitive to small uncertainties in the material parameters.

6.3 Validity of 2D Surrogates for 3D Volumetric Studies

A key objective of the study was to evaluate whether a two-dimensional model can serve as a computationally efficient surrogate for three-dimensional simulations. The results provide strong support for this approximation in rectilinear geometries.

The attenuation coefficients obtained for the two-dimensional square and the three-dimensional cube differ by only a few percent, while the calculated phase velocities are nearly identical. This close agreement indicates that the plane-strain formulation successfully captures the dominant physical mechanisms governing shear-wave propagation within the simulated domain.

Further evidence comes from the mesh sensitivity and z-direction uniformity analyses. The out-of-plane displacement variation remains negligible across the frequency range, confirming that the wave field remains effectively two-dimensional. Even at the highest tested frequency, the observed variation in the z-direction is largely attributable to numerical noise rather than meaningful physical behaviour.

These findings suggest that the 2D square model can accurately reproduce the volumetric decay characteristics of the 3D cube while requiring significantly lower computational resources. For large parametric studies or iterative model development, this reduction in complexity is particularly advantageous.

However, the parametric sensitivity analysis also reveals a limitation of this surrogate approach. While the 2D and 3D models agree well in terms of global attenuation behaviour, the fine-scale coupling between certain material properties may vary slightly between these two frameworks. This indicates that full 3D models may still be necessary when detailed material-property estimation or highly anisotropic tissue structures are under investigation.

6.4 Limitations and Future Work

Several modelling simplifications should be considered when interpreting the results of the present study. The simulations assume homogeneous and isotropic material properties, whereas real brain tissue exhibits regional heterogeneity and structural anisotropy associated with white matter fibre architecture. In addition, the simplified geometries neglect anatomical features such as sulci, ventricles, and vascular structures that locally influence wave propagation. Crucially, the idealized nature of the circular domain may introduce geometric resonance effects such as the observed energy trapping and frequency-dependent sensitivity spikes, that might be attenuated or absent in the irregular geometries of actual neuroanatomy.

Furthermore, the excitation was modelled as a purely harmonic displacement input applied at the boundary, whereas experimental elastography systems often generate more complex forcing conditions through mechanical actuators or acoustic radiation forces. These simplifications were intentionally adopted in order to isolate the effects of geometry and material parameters within a controlled computational framework. Additionally, while the 2D plane-strain approximation proved highly effective for rectilinear domains, its validity for more complex, non-rectilinear volumetric geometries remains to be established.

Future work could extend the present modelling approach by incorporating anatomically realistic brain geometries derived from medical imaging, as well as spatially heterogeneous material properties. Such extensions would allow the interaction between wave propagation, tissue structure, and regional mechanical properties to be explored in greater detail within patient-specific computational models. Further investigation into the frequency-dependent coupling between porosity and boundary geometry is also warranted to better understand the transition between intrinsic poroelastic damping and geometric scattering.

7 Conclusion

This study presented a systematic finite element investigation of shear wave propagation in brain-like poroelastic media, implemented across three benchmark geometries (a two-dimensional square, a two-dimensional circle, and a three-dimensional cube) using COMSOL Multiphysics. By independently con-

trolling geometry, boundary treatment, and material parameters, the simulations addressed the three research questions posed at the outset and established a reproducible computational framework for evaluating the assumptions underlying standard MRE wave inversion.

The results demonstrate that boundary treatment is a prerequisite for physically interpretable wave fields in all geometries. Reflective hard-wall boundaries consistently produced standing-wave interference that obscured the propagating shear-wave front, while the combination of Perfectly Matched Layers and periodic lateral boundary conditions suppressed reflected energy and yielded spatially coherent displacement profiles. For the circular geometry, an annular PML paired with a boundary-layer mesh at the domain–PML interface was required to mitigate the additional scattering introduced by the curved boundary.

With appropriate boundary treatment in place, domain geometry emerged as the primary factor differentiating wave propagation behaviour across the three models. The two-dimensional square and three-dimensional cube produced nearly identical displacement profiles, attenuation coefficients, and phase velocities at 50 Hz (approximately 1.9–2.0 $\mathrm{m\,s^{-1}}$), differing by only 5.2% in attenuation coefficient and falling within the range reported by in vivo MRE studies of human brain tissue. This close agreement validates the plane-strain approximation as a computationally efficient and physically accurate surrogate for volumetric rectilinear domains. The two-dimensional circle, by contrast, exhibited an attenuation coefficient nearly double that of the rectilinear models and a penetration depth approximately 50% shorter, attributable to geometric scattering from the curved boundary rather than to differences in intrinsic material dissipation. This finding has a direct implication for MRE inversion: circular or anatomically curved domains can produce apparent attenuation that substantially exceeds the true poroelastic value, potentially biasing stiffness estimates derived under plane-wave assumptions.

The parametric sensitivity analysis identified hydraulic permeability as the dominant material parameter governing displacement magnitude and energy dissipation across all three geometries and across the full frequency range. This result is consistent with Biot's theoretical framework, in which permeability controls the resistance to interstitial fluid flow and thereby determines the rate of viscous energy dissipation during each oscillation cycle [6]. At lower excitation frequencies, porosity exerted a secondary influence, particularly in the circular geometry, reflecting the longer oscillation periods available for fluid redistribution within the pore space. The global coefficient of variation remained below 3.2% across all geometries, confirming that the model response is robust to the parameter uncertainties tested and that the numerical implementation is stable throughout the simulated frequency range.

Taken together, these findings establish that simplified plane-wave MRE inversion assumptions are most credible in rectilinear, laterally bounded domains with adequate absorbing boundaries, where two-dimensional plane-strain models reliably reproduce the volumetric wave dynamics at a fraction of the computational cost. In geometries with curved boundaries, geometric scattering introduces systematic attenuation artefacts that cannot be attributed to tissue stiffness alone, and more sophisticated inversion strategies accounting for domain geometry would be required for accurate stiffness reconstruction. The poroelastic framework, with permeability as its key free parameter, captures the essential dissipation physics of fluid-saturated brain tissue within the clinically relevant excitation range and provides a principled basis for future studies incorporating anatomically realistic geometries and spatially heterogeneous material distributions.

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Annexes

Cube Mesh Sensitivity — z-Direction Uniformity

Figure 19 shows the displacement magnitude along a z-direction cutline through the domain centre at 20, 50, and 90 Hz. At 20 and 50 Hz variation across the two z-layers remains below 0.5%, confirming that the periodic boundary conditions on the z-faces produce a uniform field and that $z = 2$ introduces no meaningful out-of-plane error across the main frequency range of interest. At 90 Hz the apparent variation reaches approximately 9%; as discussed in Section 5.2.3, this figure corresponds to an absolute displacement of approximately 0.025 μm in the z-direction and is not physically significant.

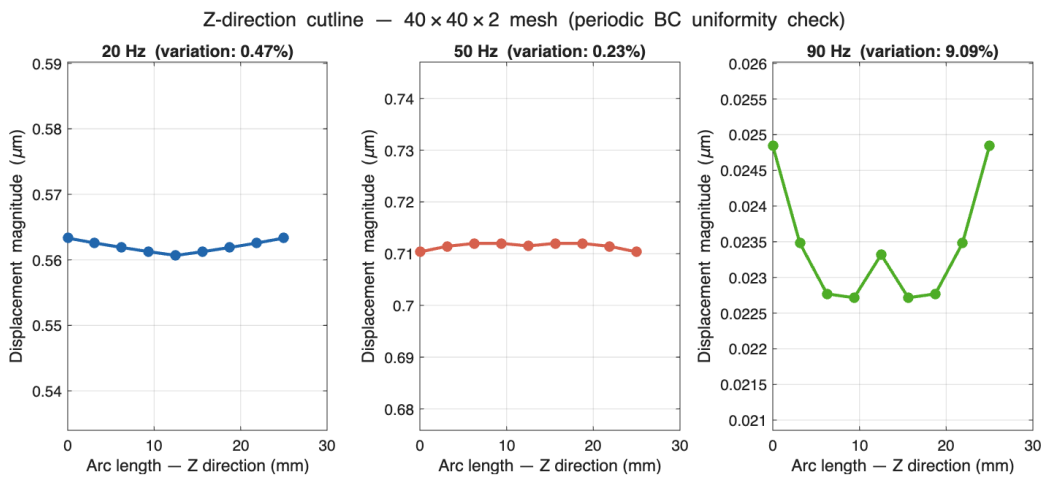

Figure 19. *Displacement magnitude along the z-direction cutline at* 20, 50, *and* 90 Hz. *Variation across the two z-layers is negligible at* 20 *and* 50 Hz. *The elevated apparent variation at* 90 Hz *corresponds to a near-zero absolute displacement and is not physically significant.*